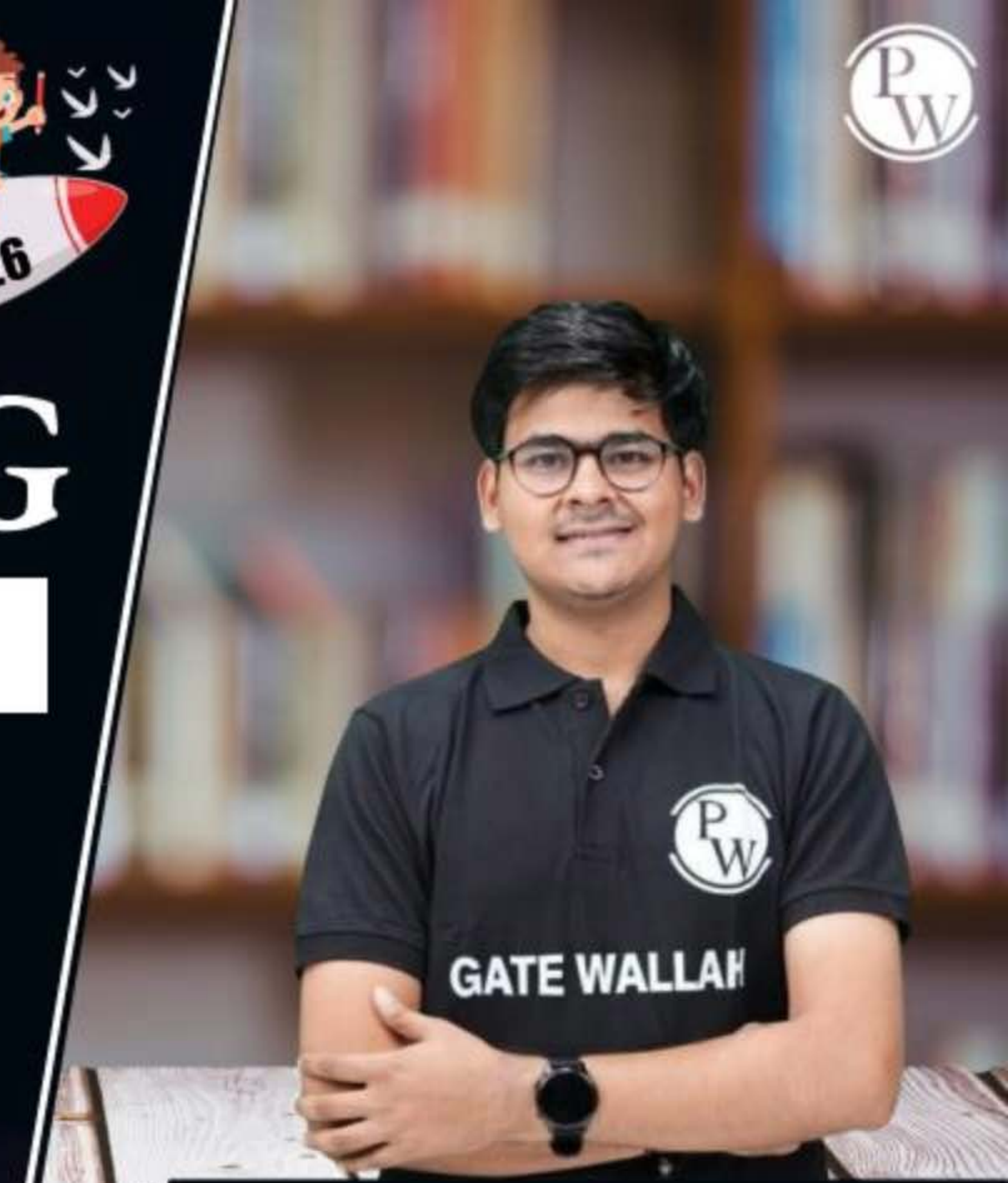
CS & IT ENGINEERING

Digital Logic

Boolean Theorems and Logical Gates

Lecture No. 3



By- Aditya sir

Recap of Previous Lecture

Topic

Redundancy Thm.

SOP

POS

H.W

Practice

Topics to be Covered

Topic

H.W Soln

SOP & POS Contd...

Logical Gates

About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.

Telegram

@ADITYASIR_PW

Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

1) $\bar{A}B + AC + \bar{B}C$

not

A) $\bar{A}B + AC$

B) $\bar{A}B + C$

C) $AC + \bar{B}C$

D) $\bar{A}B + \bar{B}C$

A ×

B ×

C ×

$\bar{A}B + AC + \bar{B}C$

$= \underset{x}{\bar{A}B} + \underset{\bar{x}}{(A+\bar{B})}\cdot C \implies X + \bar{X}\cdot C$

Let $\bar{A}B = X$

$\overline{\bar{A}B} = \bar{X}$

$(A+\bar{B}) = \bar{X}$

De Morgan's Thm. $\overline{\bar{A}B} = (A+\bar{B})$

$X+\bar{X}C = (X+\bar{X})(X+C)$

$= 1\cdot(X+C) = X+C = \bar{A}B + C$

Property :- $\boxed{A \cdot (A + \text{anything}) = A}$

Proof :-

$$= A \cdot A + A \cdot \text{anything}$$

$$= A + A \cdot \text{anything}$$

$$= A(1 + \text{anything})$$

$$= A \cdot 1 = A$$

2) $\overline{A\bar{B}(\bar{A}B + \bar{B}C + A\bar{B}D + A\bar{B}\bar{D})}$

$= \overline{A\bar{B}\left[\bar{A}B + \bar{B}C + A\bar{B}(D+\bar{D})\right]}$

$= \overline{A\bar{B}\left[A\bar{B} + \bar{A}B + \bar{B}C\right]} = \overline{A\bar{B}} = \bar{A} + \bar{\bar{B}}$

$= X\left[X + \text{anything}\right]$

$= (\bar{A} + B)$ ✓

3) $A + BC + \bar{A}C$

$= A + \bar{A}C + BC$

$= (A + \bar{A})(A + C) + BC$

$= 1 \cdot (A + C) + BC$

$= A + C + BC$

$A + C + CB$

$= A + C(1 + B)$

$= A + C \cdot 1$

$= \underline{\underline{A + C}}$

4) $\overline{(\bar{A}+\bar{B})\cdot(\bar{B}+\bar{C})}$

$= \overline{(\bar{A}+\bar{B})} + \overline{(\bar{B}+\bar{C})}$

$= A\cdot B + B\cdot C$

$= \underline{\underline{B(A+C)}}$

$\rightarrow$

$\overline{(\bar{B}+\bar{A})(\bar{B}+\bar{C})}$

$= \overline{\bar{B}+\bar{A}\,\bar{C}}$

$= B\cdot\overline{(\bar{A}\cdot\bar{C})}$

$= \underline{\underline{B\cdot(A+C)}}$

5) $\bar{A}\bar{B} + AC + \bar{B}C = ?$

redundant

A ✓

B ✗

C ✗

$$\boxed{\bar{A}\bar{B} + AC}$$

(Q) $\overline{\boxed{\bar{P}Q}R\left[PQ+\bar{Q}R+\bar{P}Q+QRS+\bar{P}RS\right]}$

$= \overline{R\cdot\bar{P}\cdot Q\left[\bar{P}Q+\text{remaining}\ldots\right]}$

$= \overline{R\cdot\bar{P}Q} = \left(\bar{R}+P+\bar{Q}\right)$

(Q) $\overline{\boxed{\bar{P}Q}R\left[PQ+\bar{Q}R+\bar{P}Q+QRS+\bar{P}RS\right]}$

$\Downarrow$

$\overline{\bar{P}QR\left[Q(P+\bar{P})+\bar{Q}R+QRS+\bar{P}RS\right]}$

$= \overline{\bar{P}QR\left[\cancel{(Q+\bar{Q})}(Q+R)+QRS+\bar{P}RS\right]}$

$= \overline{\bar{P}R\cdot Q\left[Q+R+QRS+\bar{P}RS\right]}$

$= \overline{\bar{P}RQ} = P+\bar{R}+\bar{Q}$

$$1 \qquad\qquad 0$$

$$f(P,Q) = \sum(1,3) = \prod(0,2)$$

$$\bar{f}(P,Q) = f_1(P,Q) = \prod(1,3) = \sum(0,2)$$

eg- $F(P,Q,R) = \underline{\underline{\sum(1,2,5)}}$

$\overline{F}(P,Q,R) = \prod(1,2,5) = \underline{\underline{\sum(0,3,4,6,7)}}$

$\underbrace{PQR}$

$\underline{\underline{0-7}}$

eg2) $F(P,Q,R) = \underline{\Pi(0,2,6)}$

$\bar{F}(P,Q,R) = \Sigma(0,2,6) = \Pi(1,3,4,5,7)$

Basic Gates	Arithmatic Gates	Universal Gates
→ AND	→ XOR	→ NAND
→ OR	→ XNOR	→ NOR
→ NOT		

✓

1) OR Gate

Repr :

$\Downarrow$

$A + B$

Truth Table

	A	B	y
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	1

SOP (1s)

$A \to 1$
$\bar{A} \to 0$

$\Sigma(1,2,3)$

1 2 3

01 10 11

$(\bar{A}B) + (A\bar{B}) + (AB)$

$A\bar{B} + AB + \bar{A}B$

$A(B+\bar{B}) + \bar{A}B = A\cdot 1 + \bar{A}B = A + \bar{A}B = (A+B)(A+\bar{A})$

$= (A+B)$

$\underbrace{B+\bar{B}}_{1}$

POS (0s)

$A \to 0$
$\bar{A} \to 1$

$\Pi(0)$

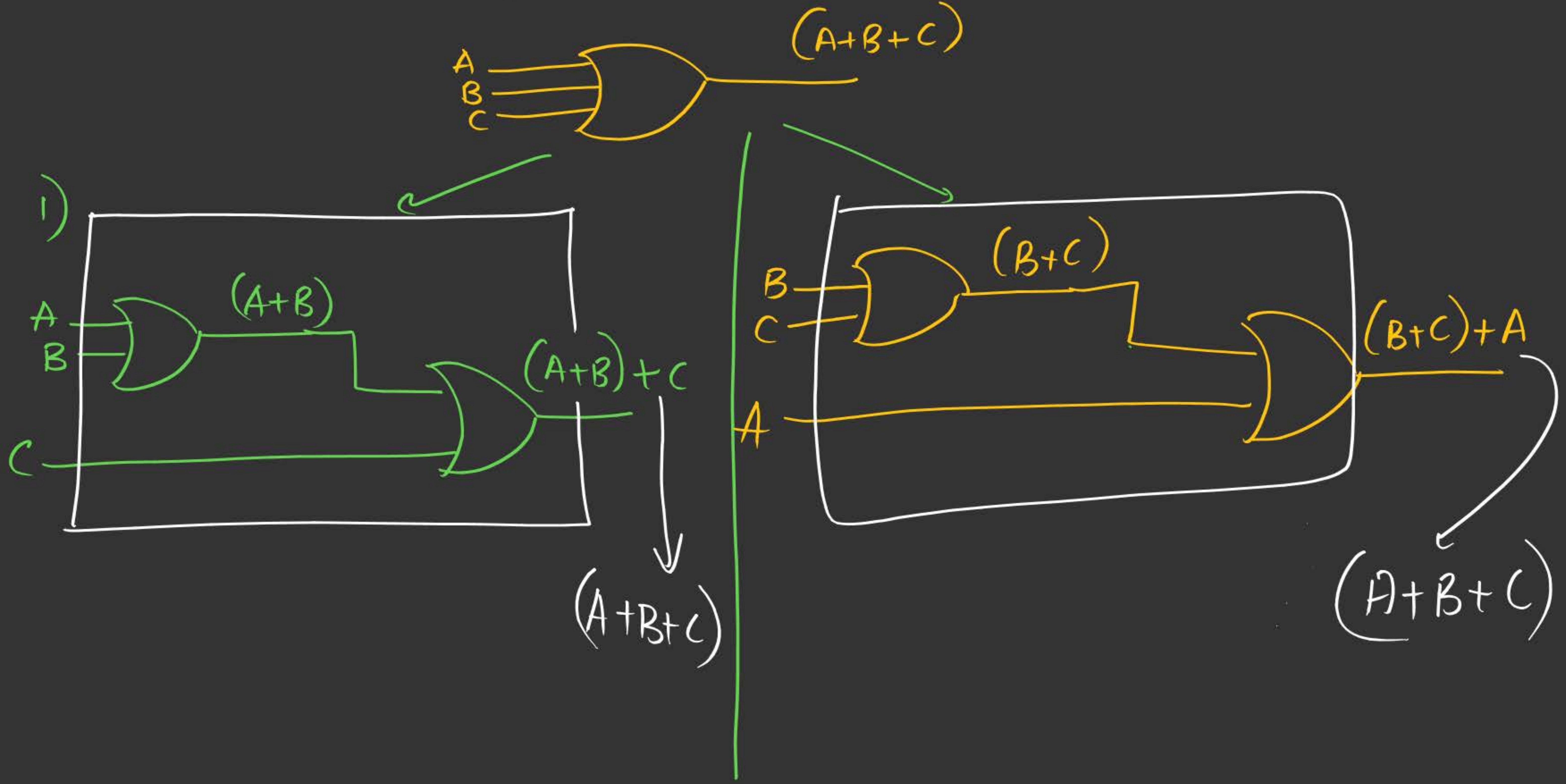
$(A+B)$

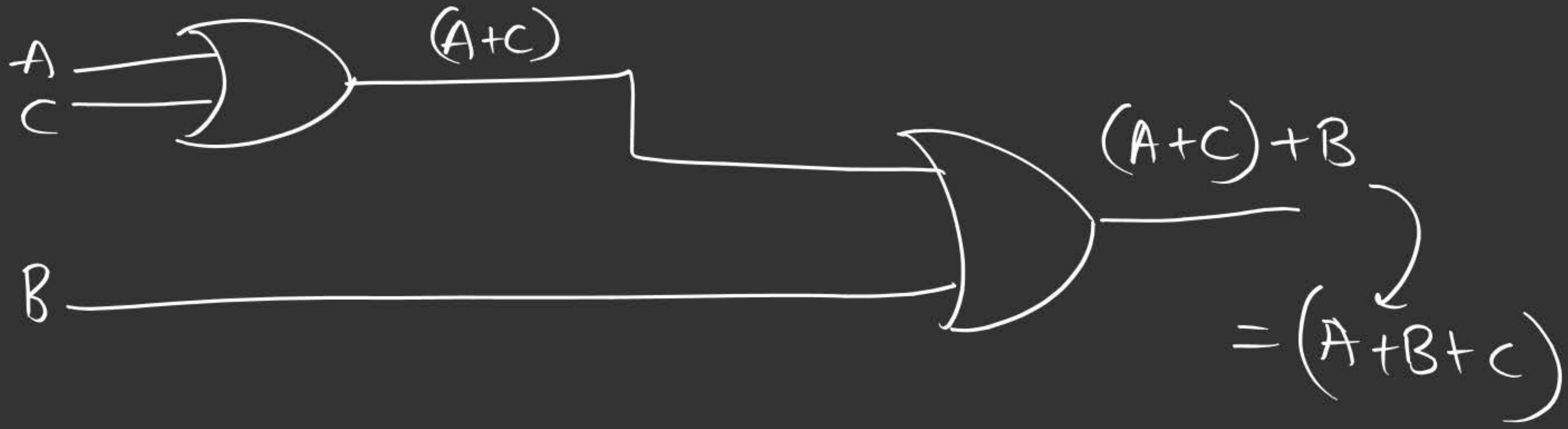
1) Commutative Law

$$\boxed{(A+B) = (B+A)}$$

OR Holds Commutative Law

2) Associative law




A
C
(A+C)
B
(A+C)+B
=(A+B+C)

→ OR holds associative law.

<u>Application:</u>

Multiple i/p's OR gate can be implemented using 2 i/p's OR gate.

* Enable and Disable i/p

1) Enable

$$A+B+0=(A+B)$$

Control i/p = 0

2) Disable

$$(A+B)+1=1$$

Control i/p = 1

0	0
0	1
1	0
1	1

$\Big\}$ 1

Imp Obs :-

1> if even one i/p line is 1, O/p always 1.

2> The O/p is 0 only when all the i/p's are 0.

2) AND Gate :

Repr :-

$$y = (A \cdot B)$$

Truth table

	A	B	y
0	0	0	0
1	0	1	0
2	1	0	0
3	1	1	1

$A \rightarrow 1$
$\bar{A} \rightarrow 0$

SOP (1s)

$\Sigma(3)$

$\downarrow$

11

$(A \cdot B)$

$A \rightarrow 0$
$\bar{A} \rightarrow 1$

POS (0s)

$\Pi(0,1,2)$

00   01   10

$(A+B) \cdot (A+\bar{B}) \cdot (\bar{A}+B)$

$[A + (B \cdot \bar{B})] \cdot (\bar{A}+B)$

$(A+0) \cdot (\bar{A}+B) = A(\bar{A}+B)$

$= A \cdot \bar{A} + A \cdot B$

$= 0 + AB = AB$

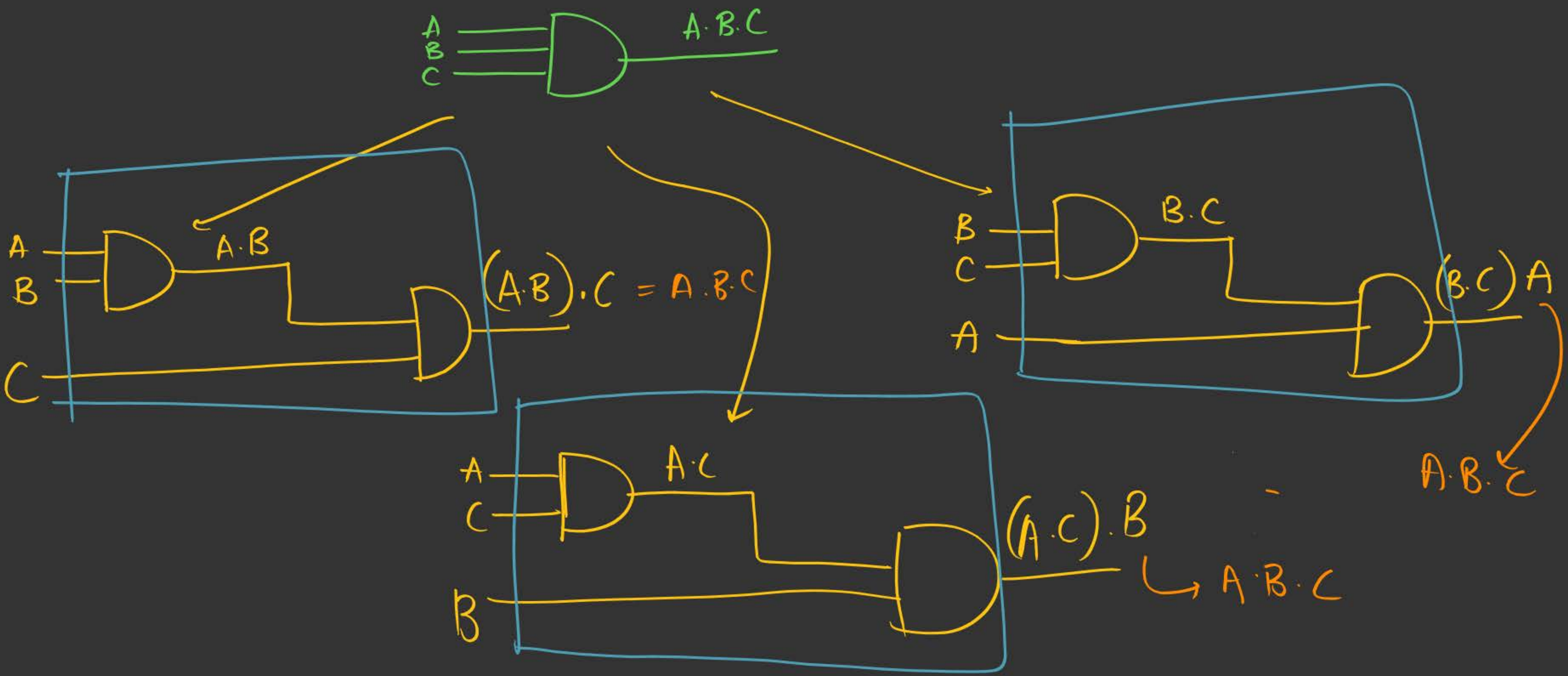
1) Commutative law :-

$$(A \cdot B) = (B \cdot A)$$

(order of i/p doesnt matter)

"AND holds Commutative law"

2) Associative Law:



Appl: Multiple i/p's AND gate

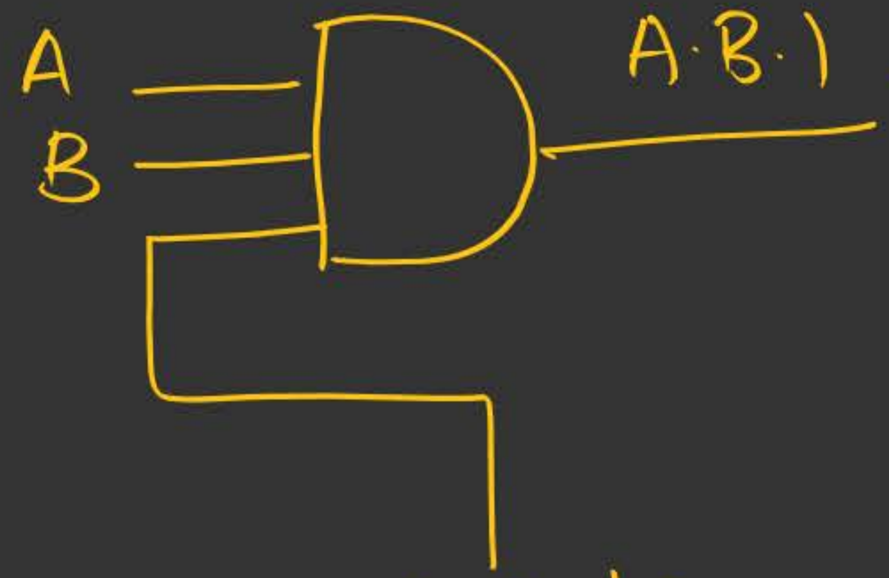
Can be implemented using 2 i/p's AND gate

AND holds association property/law

* Enable & Disable i/p for AND.

1) Enable :

$A \cdot B \cdot 1 = A \cdot B$



control i/p = 1

2) disable :

$A \cdot B \cdot 0 = 0$

control i/p = 0

$$\left.\begin{array}{cc} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{array}\right\} 0$$

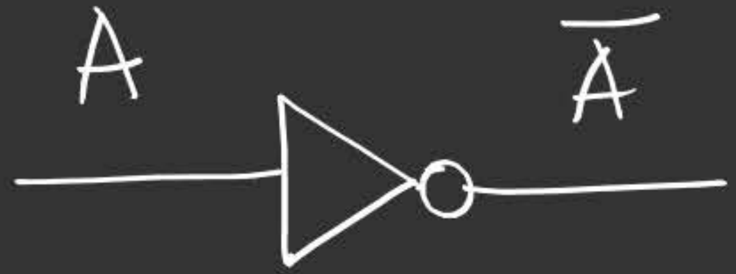
Imp obs :-

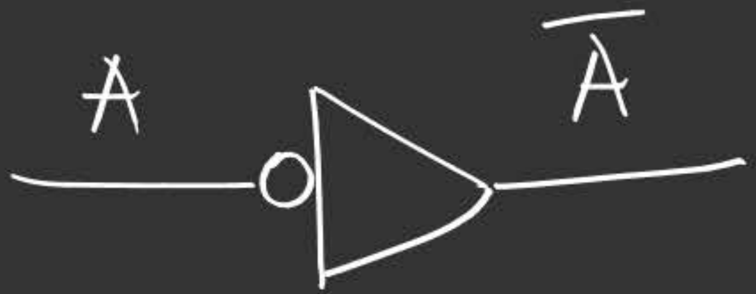
1) If any i/p line is 0, the o/p will be 0.

2) The o/p is 1 only when all the i/p's are 1.

3) NOT gate

Repr :





Truth table

	A	$y = \overline{A}$
0	0	1
1	1	0

$A \to 1$

$\overline{A} \to 0$

SOP

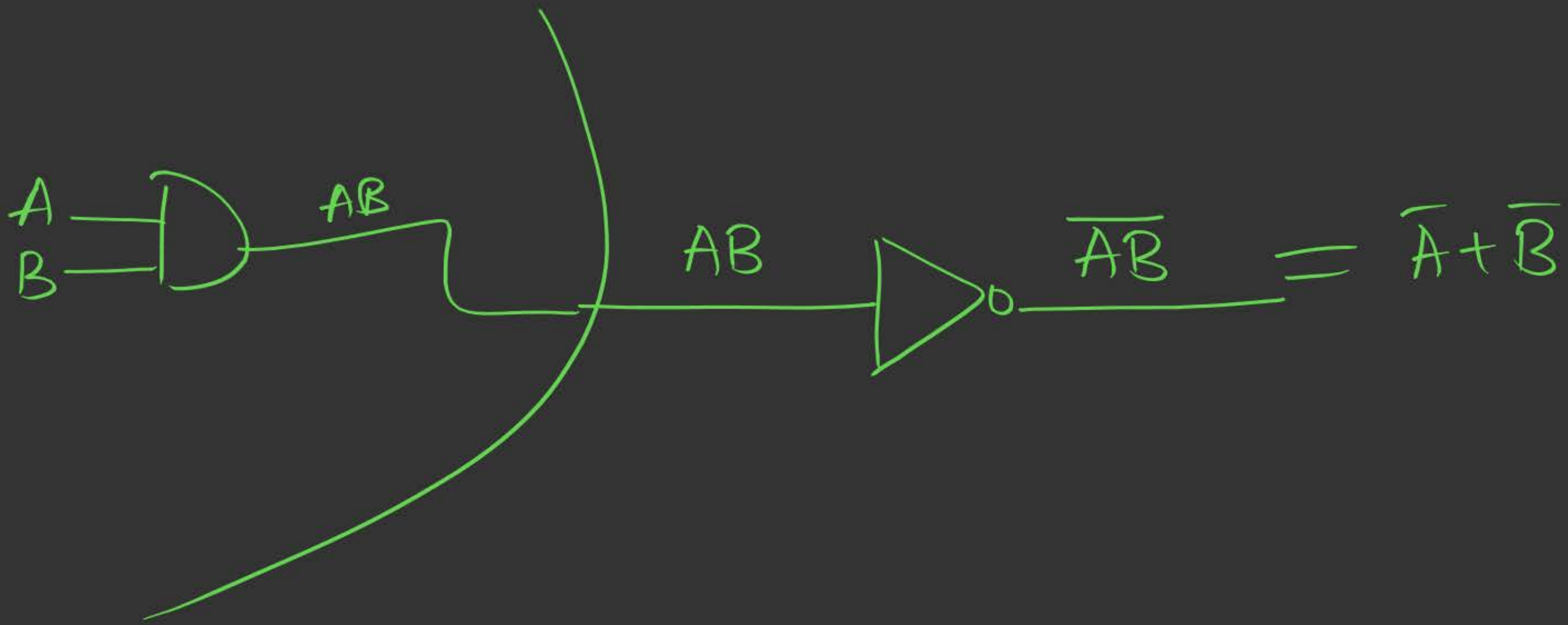
$$\Sigma(0)$$

$$= \overline{A}$$

POS

$$\Pi(1)$$

$$= \overline{A}$$



A
B
AB
AB
$\overline{AB}$
$= \bar{A} + \bar{B}$

2 mins Summary

Topic

Topic

Topic

Topic

Topic

H.W Soln

Basic gates

THANK - YOU